

Shift11

AI-Powered Football Momentum Attribution

Microsoft-First Learning Resources

Scope

Resources are limited to what is required for Shift11: Python foundations, data/file handling, Git/GitHub, deterministic analytics, Microsoft Agent Framework, Microsoft Foundry and practical Azure deployment. Advanced ML, RAG, vector databases, Kubernetes and unrelated cloud topics are intentionally excluded.

Phase 1

Python Fundamentals

<https://learn.microsoft.com/en-us/training/paths/get-started-with-python-fundamentals/>

Beginner Python foundations.

Phase 1

Python Building Programs

<https://learn.microsoft.com/en-us/training/paths/get-started-with-python-programs/>

Loops, collections, files and reusable code.

Phase 1

Python Collections

<https://learn.microsoft.com/en-us/training/modules/organize-data-with-collections-python/>

Lists/dictionaries for event and fact models.

Phase 1

Python File I/O

<https://learn.microsoft.com/en-us/training/modules/read-write-files-python/>

JSON/file handling.

Phase 1

Reusable Python Code

<https://learn.microsoft.com/en-us/training/modules/write-reusable-code-python/>

Functions and modular structure.

Phase 1

VS Code + Python

<https://learn.microsoft.com/en-us/training/modules/install-configure-visual-studio-code-python/>

Environment, pip and VS Code workflow.

Phase 1

Introduction to Git

<https://learn.microsoft.com/en-us/training/modules/intro-to-git/>

Version control fundamentals.

Phase 1

GitHub Foundations

<https://learn.microsoft.com/en-us/training/paths/github-foundations/>

Repositories and GitHub workflow.

Phase 5

Microsoft Agent Framework Overview

<https://learn.microsoft.com/en-us/agent-framework/overview/?pivots=programming-language-python>

Agents, workflows and Python framework.

Phase 5

Agent Framework Documentation

<https://learn.microsoft.com/en-us/agent-framework/>

First agent, tools, structured output, workflows and hosting.

Phase 5

First Agent

<https://learn.microsoft.com/en-us/agent-framework/get-started/your-first-agent?pivots=programming-language-python>

Build the first Python agent.

Phase 5

Microsoft Foundry Overview

<https://learn.microsoft.com/en-us/azure/ai-foundry/what-is-foundry?view=foundry>

Foundry developer surfaces and hosted-agent concepts.

Phase 5

Foundry SDKs and Endpoints

<https://learn.microsoft.com/en-us/azure/foundry/how-to/develop/sdk-overview?pivots=programming-language-python%2F%2F>

Python SDK and Agent Framework positioning.

Phase 8

Foundry Hosted Agents

<https://learn.microsoft.com/en-us/azure/foundry/how-to/develop/framework-hosted-agents>

Deploy Agent Framework agents as hosted agents.

Phase 8

Hosted Agent Quickstart

<https://learn.microsoft.com/en-us/azure/ai-foundry/agents/quickstarts/quickstart-hosted-agent>

Practical deployment path.

Phase 8

Azure Container Apps

<https://learn.microsoft.com/en-us/azure/container-apps/get-started>

Container deployment if required.

Recommended Learning Order

1. Python fundamentals → collections → files → reusable code.
2. VS Code, virtual environments and pip.
3. Git/GitHub.
4. Build the synthetic event generator.
5. Learn only the Python needed for momentum and attribution.
6. Learn Agent Framework: first agent, tools/structured output, then only required workflows/hosting.
7. Learn Foundry once the local fact-pack → explanation pipeline works.
8. Learn Azure deployment only after the local demo is stable.

Microsoft-First Stack

Microsoft's current documentation describes Agent Framework as a Python-capable orchestration framework and identifies it as a recommended orchestration layer for Foundry hosted agents. Foundry supports code-based hosted agents with managed runtime capabilities. Use these resources only as they become relevant to the corresponding project phase.

Do Not Over-Learn

For the hackathon, do not make advanced machine learning, RAG, vector databases, Kubernetes, complex MLOps or multi-agent architectures prerequisites. Shift11's core value is deterministic momentum detection, player attribution, evidence chains and grounded AI explanation.